



Family Letter — Unit 1

Intermediate Algebra — Unit 1: Variables and Expressions

1 FAMILY LETTER

Dear Families, Welcome to Intermediate Algebra! We're excited to begin a year of mathematical discovery with your student.

What Your Student Will Learn in Unit 1

This first unit builds the foundation for all of algebra. Students will:

1. **Use variables** to represent unknown quantities
2. **Write expressions** from verbal descriptions
3. **Apply GEMDAS** (the order of operations) to evaluate expressions accurately
4. **Substitute values** for variables and simplify
5. **Combine like terms** and use the distributive property
6. **Apply algebra to real-world financial situations** — like calculating discounts and sales tax

**Standards covered: 9.3.5.1, 9.3.5.3, 9.3.5.5, 9.3.5.7, 9.3.5.8.

Unit 1 Rubric: How Grades Are Earned

Grades are earned on a locked 4-level scale: **Emerging** → **Developing** → **Proficient** → **Superlative**. **Proficient is the passing level; Superlative earns an A.**

Benchmark	Emerging	Developing	Proficient	Superlative
9.3.5.1	Recognizes but cannot yet apply scientific notation.	Applies scientific notation with support or on routine problems only.	Correctly applies scientific notation to solve problems and explain reasoning.	Independently applies scientific notation to novel problems and justifies the most efficient approach.
9.3.5.3	Recognizes but cannot yet apply number systems.	Applies number systems with support or on routine problems only.	Correctly applies number systems to solve problems and explain reasoning.	Independently applies number systems to novel problems and

Benchmark	Emerging	Developing	Proficient	Superlative
				justifies the most efficient approach.
9.3.5.5	Recognizes but cannot yet apply cost, discounts, taxes & order of operations.	Applies cost, discounts, taxes & order of operations with support or on routine problems only.	Correctly applies cost, discounts, taxes & order of operations to solve problems and explain reasoning.	Independently applies cost, discounts, taxes & order of operations to novel problems and justifies the most efficient approach.
9.3.5.7	Recognizes but cannot yet apply equivalent forms & rearranging formulas.	Applies equivalent forms & rearranging formulas with support or on routine problems only.	Correctly applies equivalent forms & rearranging formulas to solve problems and explain reasoning.	Independently applies equivalent forms & rearranging formulas to novel problems and justifies the most efficient approach.
9.3.5.8	Recognizes but cannot yet apply writing expressions in multiple ways.	Applies writing expressions in multiple ways with support or on routine problems only.	Correctly applies writing expressions in multiple ways to solve problems and explain reasoning.	Independently applies writing expressions in multiple ways to novel problems and justifies the most efficient approach.

Key Vocabulary You Might See

Term	Simple Definition
Variable	A letter that stands for an unknown number
Expression	A math phrase with numbers, variables, and operations
Coefficient	The number in front of a variable
GEMDAS	Grouping → Exponents → Multiply/Divide → Add/Subtract
Evaluate	Find the value of an expression
Like terms	Terms with the same variable part (like $3x$ and $5x$)

How You Can Help at Home

- Ask your student: **“What does GEMDAS stand for?”** (Grouping, Exponents, Multiplication/Division, Addition/Subtraction) • Have your student explain why $3 + 4 \times 2 = 11$ but $(3 + 4) \times 2 = 14$ • Talk about math in everyday life: **“If we buy 3 notebooks for \$ 2 each and use a \$ 5 coupon, how would we write that as an expression?”** • Ask about the **Merit Time** games they’ll play in class — starting with **Translation Match** and **Expression Race** in the first lessons

Merit Time PBIS Games

Unit 1 features five brand-new **Merit Time games** — **Translation Match**, **Expression Race**, **GEMDAS Challenge**, **Code Match**, and **Speed Lane** — one introduced in each of the first five lessons. These zero-device, team-based games build fluency while practicing translating phrases, GEMDAS, and evaluating expressions. Students play these games in the first five lessons to build fluency.

MN Companies in Our Problems

Students will see problems connected to Minnesota companies including **Target** (Minneapolis), **3M** (Maplewood), **Mayo Clinic** (Rochester), **Xcel Energy** (Minneapolis), and **U.S. Bancorp** (Bloomington). These real-world connections show how algebra is used every day in our own community.

Cultural Connections

Our curriculum honors Minnesota's Native American heritage per **MN 120B.021**. Unit 1 incorporates **Ojibwe and Dakota** cultural contexts in mathematics, including wild rice harvesting (using expressions to track yields) and Dakota star knowledge (moon cycles and seasonal movement).

Assessment

Students will take two short quizzes (after Lessons 5 and 15) and an end-of-unit test. All assessments are tagged to Minnesota state standards, and results will be available through the school's learning management system.

Questions?

If you have questions about the unit, please reach out to your student's teacher. Thank you for partnering with us in your student's mathematical journey! **Merit Education Group**™
